

Executive Summary – Customer Churn Analysis

This analysis examined customer demographics, service usage patterns, contract types, tenure, and payment behaviors to identify the key drivers of customer churn. Data quality issues were addressed by cleaning and converting the **TotalCharges** field and transforming categorical variables to improve interpretability.

Key Findings

- **Overall Churn Rate:** Approximately **26.5% of customers have churned**, indicating that nearly one in four customers discontinue the service.
- **Customer Tenure:** Customers with **shorter tenures (1–2 months)** exhibit significantly higher churn, while long-term customers tend to remain loyal.
- **Senior Citizens:** The proportion of churn is higher among **senior citizens** compared to non-senior customers, suggesting this segment may require targeted retention efforts.
- **Contract Type:** Customers on **month-to-month contracts** show the highest churn rates, whereas those with **one-year and two-year contracts** are much more likely to stay.
- **Internet Service:** Customers using **Fiber Optic Internet** demonstrate noticeably higher churn compared to DSL users, making this segment a potential area for service quality and pricing review.
- **Value-Added Services:** Customers without **Online Security, Online Backup, Device Protection, and Tech Support** are considerably more likely to churn. Adoption of these services appears to improve customer retention.
- **Streaming Services:** Streaming TV and Streaming Movies do not show a strong relationship with churn when compared to security and support-related services.
- **Payment Method:** Customers paying through **Electronic Check** exhibit the highest churn levels, indicating possible dissatisfaction, payment friction, or differences in customer profiles.
- **Gender:** Churn behavior is relatively similar across genders, suggesting gender is not a major factor influencing customer attrition.

Business Recommendations

1. Focus retention campaigns on **new customers during their first few months of service**.
2. Encourage migration from **month-to-month contracts** to longer-term plans through incentives and loyalty benefits.

3. Increase adoption of **security, backup, and technical support services**, which are associated with lower churn.
4. Investigate the high churn among **Fiber Optic** customers to identify potential service, pricing, or customer experience issues.
5. Review the customer journey of users paying via **Electronic Check** and introduce targeted engagement strategies to reduce attrition.

Overall, the analysis indicates that **contract commitment, customer tenure, support/security services, internet service type, and payment method** are the strongest factors influencing customer churn, providing clear opportunities for improving customer retention and long-term revenue.